MATHS MANTHAN
• TUITION CLASSES •
e-mail: contact@mathsmanthan.co.in | Helpline: +91-8171546558

Subject: Mathematics | **Periodic Test Paper 16 (Cumulative)** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

General Instructions:

- This question paper consists of **12 questions** divided into **Five Sections**: A, B, C, D, and E.
- **Section A** comprises **5 questions** of **1 mark** each (Q1–Q4 MCQs, Q5 Assertion-Reason).
- **Section B** comprises **2 questions** (Q6–Q7) of **2 marks** each.
- **Section C** comprises **2 questions** (Q8–Q9) of **3 marks** each.
- **Section D** comprises **2 questions** (Q10–Q11) of **5 marks** each.
- **Section E** comprises **1 Applied Case Study** (Q12) of **5 marks** with internal choice in sub-part (iii).
- All questions are compulsory. Use of calculators is strictly prohibited.

SECTION A: Multiple Choice & Assertion-Reason Questions (5 × 1 = 5 Marks)

Q1. Which term of the AP $24, 21, 18, 15, \ldots$ is the first negative term? [1]

(A) 8th (B) 9th (C) 10th (D) 11th

Q2. If 7th term of an AP is 4 and its common difference is -4, its first term is: [1]

(A) 16 (B) 20 (C) 24 (D) 28

Q3. In an AP, if $a = 1$, $a_n = 20$, and $S_n = 399$, then the value of n is: [1]

(A) 19 (B) 21 (C) 38 (D) 42

Q4. The sum of all two-digit odd natural numbers is: [1]

(A) 2475 (B) 2530 (C) 4900 (D) 5050

Q5. **Assertion (A):** If the angles of a triangle are in AP, then one of the angles must be $60°$.
Reason (R): For angles $a - d, a, a + d$, their sum is $3a = 180° \implies a = 60°$. [1]

(A) Both (A) and (R) are true, and (R) is correct explanation.

(B) Both (A) and (R) are true, but (R) is not correct explanation.

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

SECTION B: Very Short Answer Questions (2 × 2 = 4 Marks)

Q6. Find the number of terms of the AP $18, 15\frac{1}{2}, 13, \ldots, -47$. [2]

Q7. How many terms of the AP $17, 15, 13, \ldots$ must be added to obtain a sum of 72? Explain the double answer. [2]

SECTION C: Short Answer Questions (2 × 3 = 6 Marks)

Q8. If the nth term of an AP is given by $a_n = 2n + 1$, find the sum of its first n terms. [3]

Q9. The 17th term of an AP exceeds its 10th term by 7. Find the common difference. [3]

SECTION D: Long Answer / HOTS Questions (2 × 5 = 10 Marks)

Q10. If S_1, S_2, S_3 are the sums of first n terms of three APs whose first terms are each 1 and common differences are 1, 2, and 3 respectively, prove that:

$$S_1 + S_3 = 2S_2$$

[5]

Q11. Solve for x:

$$2 + 5 + 8 + 11 + \cdots + x = 345$$

[5]

SECTION E: Applied Case Study (1 × 5 = 5 Marks)

Q12. CASE STUDY: City Auditorium Staggered Seating Capacity



A state-of-the-art civic auditorium is engineered with curved rows of tiered seating to optimize acoustics and sightlines.
The first row nearest to the performance stage contains 20 seats. Each successive row has 2 seats more than the row immediately in front of it. The auditorium contains 25 seating rows in total.

Based on the above architectural design, answer the following questions:

(i) Determine the number of seats installed in the 12th row. [1]

(ii) How many seats are there in the last (25th) row? [1]

(iii) Calculate the total seating capacity of the entire auditorium. [3]

OR

(iv) If admission tickets are priced uniformly at Rs. 300 per seat, calculate the gross ticket revenue when the auditorium is at 100% capacity. [3]

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